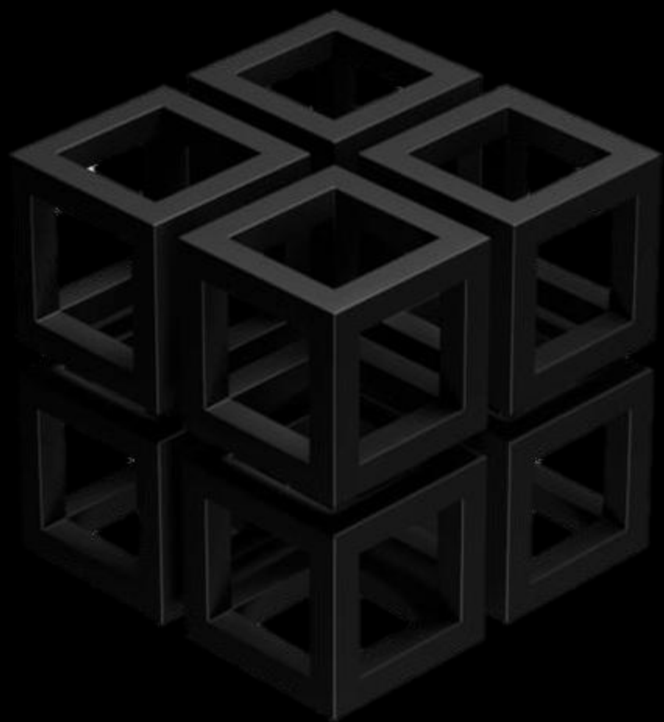

# GPX10Pro I2s-tmbuffer-aon-ww-cnn

Ambient Scientific

Jan 2026

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# Document Change History Log

| Version No. | Date | Author / Editor | Details of Change |
|---|---|---|---|
| 1.0 | 20/01/2026 | Ambient Scientific | First version |

# CONTENTS

# 1 Introduction

## 1.1 Purpose & Scope

This document describes “i2s-tmbuffer-aon-alexa-ww-cnn” application details. It is reference document for engineering, marketing team & AI’s Customers.

## 1.2 Open Points

- None

## 1.3 Definitions, Acronyms & Abbreviations

Definitions:

Acronyms:

Ambient Scientific: - AS

S9 Chip set: - AI chip set

Abbreviations:

AI: - Artificial Intelligence

BT: - Bluetooth

HW: - Hardware

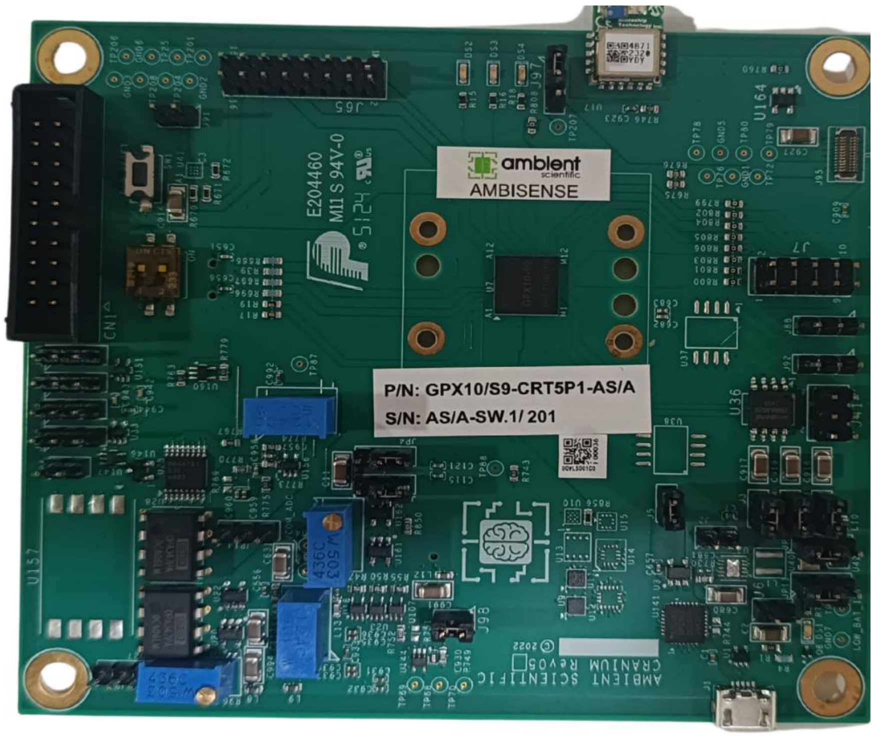
SDK: - Software Development Kit

SW: - Software

# 2 System

## 2.1 Hardware

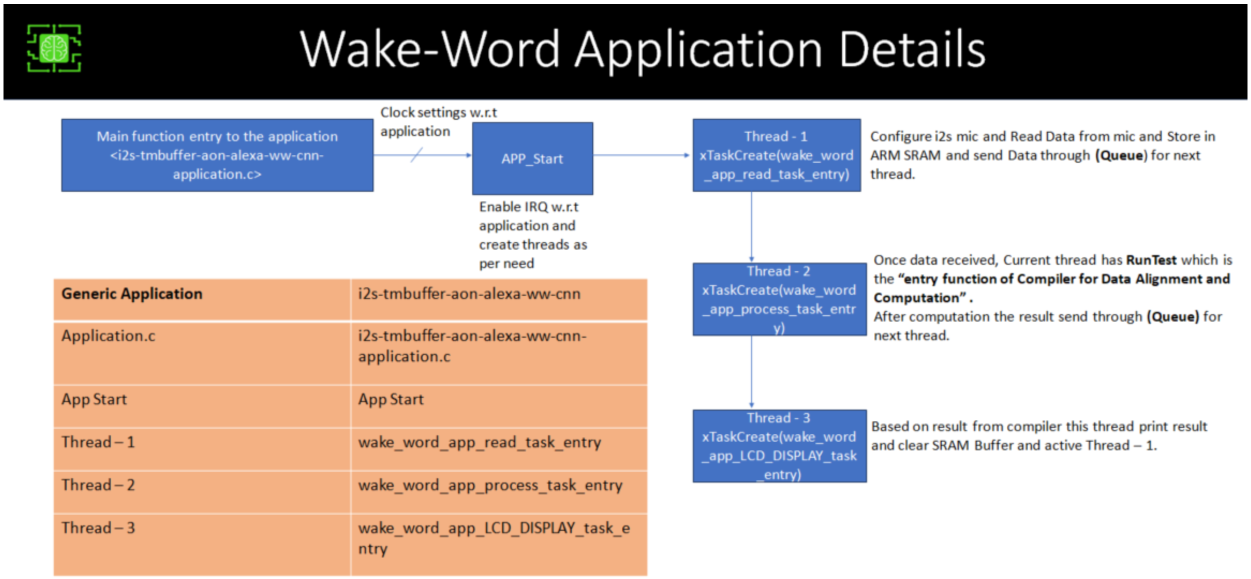
- DVK board on with I2S mic.
- JTAG and POWER cable for connection of DVK board.




## 2.2 Software

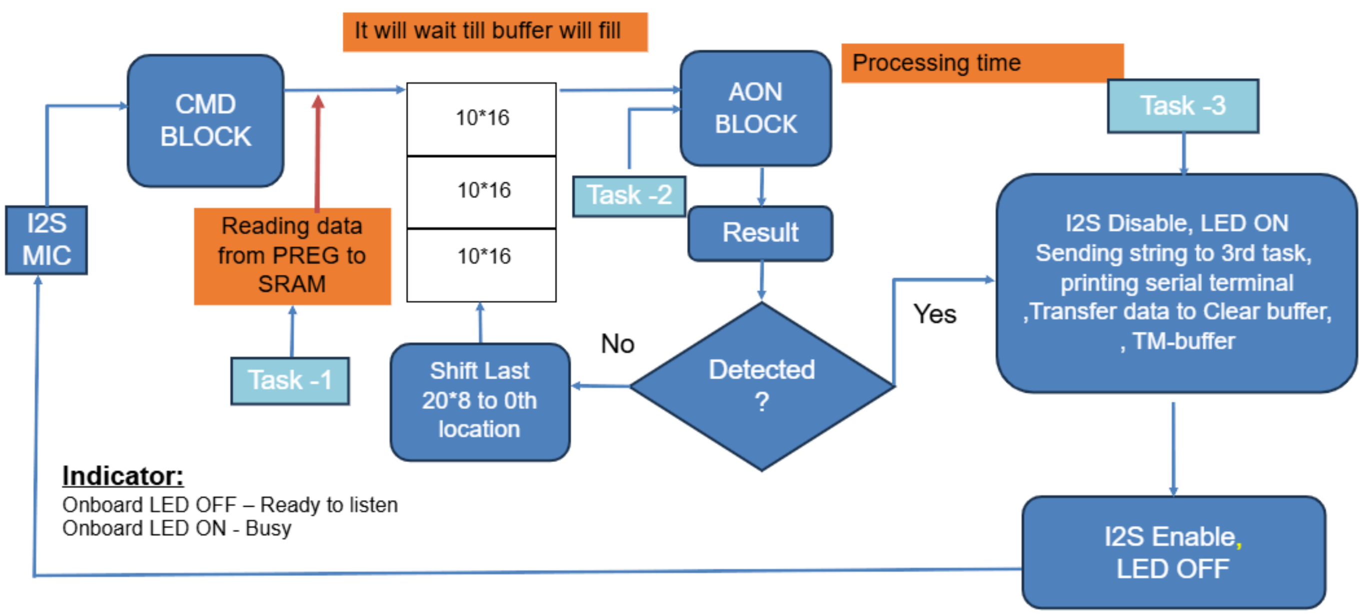
- Eclipse IDE
- Command prompt
- GPX10 PRO compiler

## 2.3 Application Pipeline




| Generic Application | i2s-tmbuffer-aon-alexa-ww-cnn |
|---|---|
| Application.c | i2s-tmbuffer-aon-alexa-ww-cnn-application.c |
| App Start | App Start |
| Thread – 1 | wake_word_app_read_task_entry |
| Thread – 2 | wake_word_app_process_task_entry |
| Thread – 3 | wake_word_app_LCD_DISPLAY_task_entry |

## 2.4 Application Flow



# 3 Loading Workspace

Please refer to “GPX10Pro_General_Application_notes” this document in this path “C:\AmbientScientific\setup\chip\fw\applications”

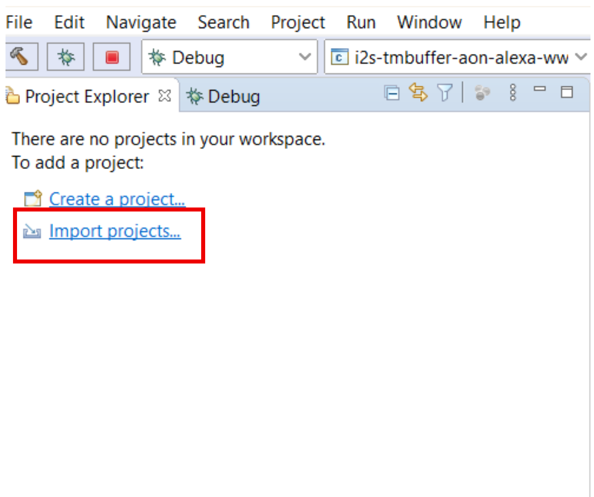
# 4 Application

## 4.1 Description:

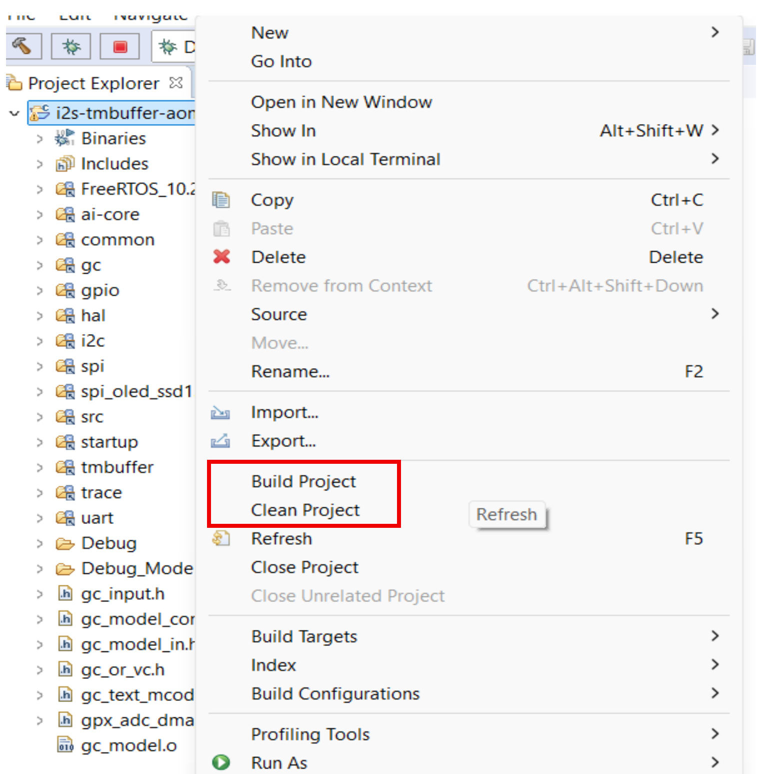
- The application involves detecting the wake-word spoken by user. It captures speaking word by I2S mic and detects the word and display the same on serial console.
- LED (GPIO_PIN_1) will indicate I2S Mic ON.
- 30*16 features generated by CMD block from 1 sec data of I2S Mic.
- 30*16 is input to the model which will give the result.
- Input: - Speak the word in front of I2S mic.
- Output: - Console prints (KEYWORD DETECTED) on serial consol.
- The application runs by eclipse and Command prompt. By default, the application runs Live mode.

## 4.2 Execution and Run

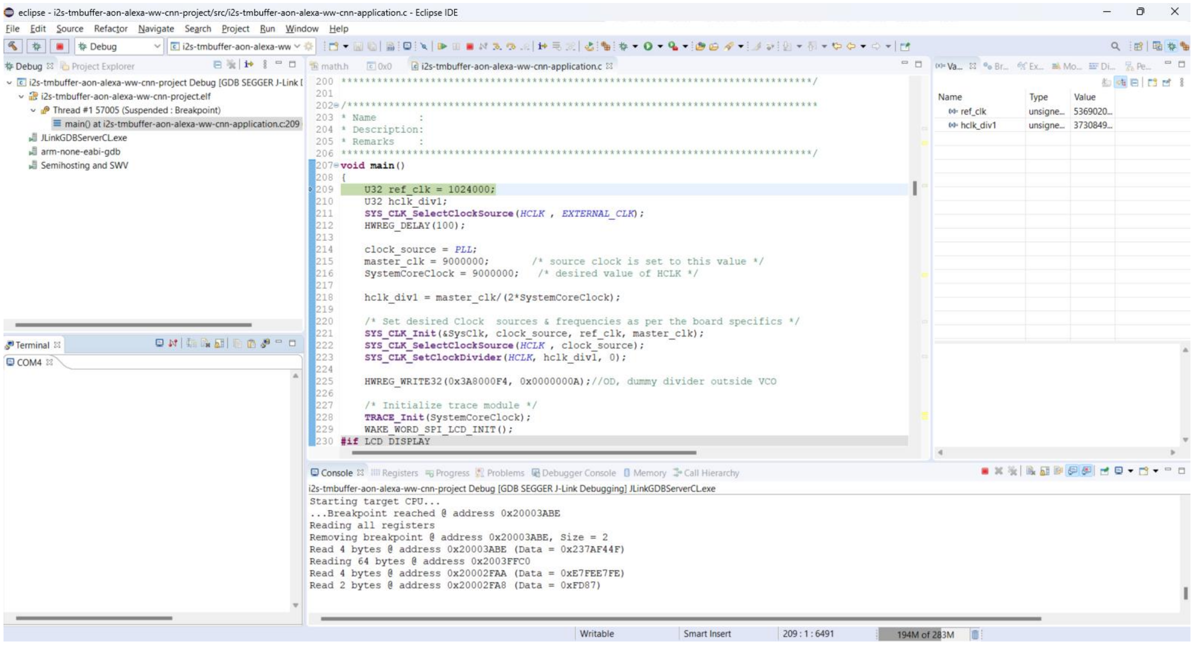
- Import application from here: "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn"



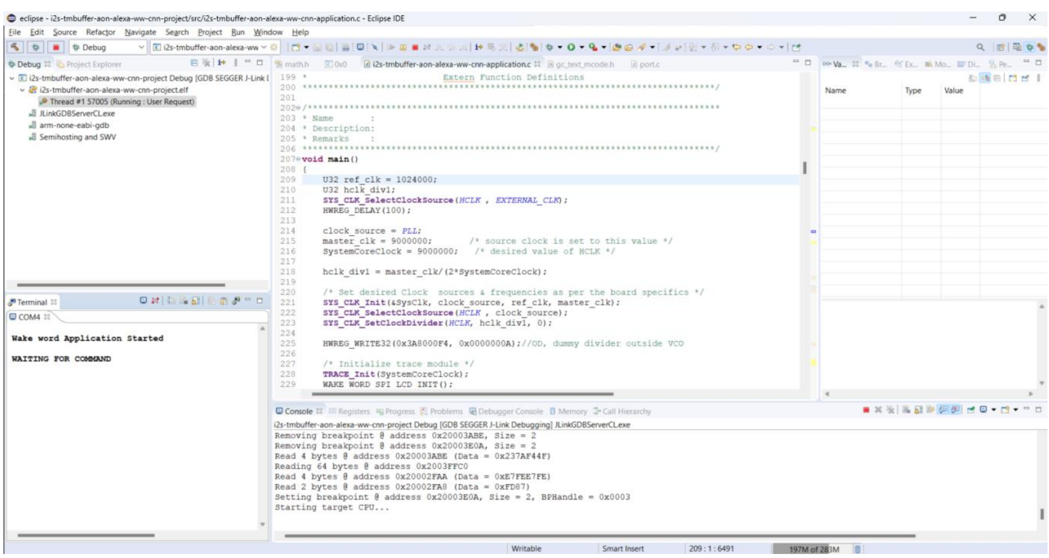

- Right click on project for **Clean and Build**



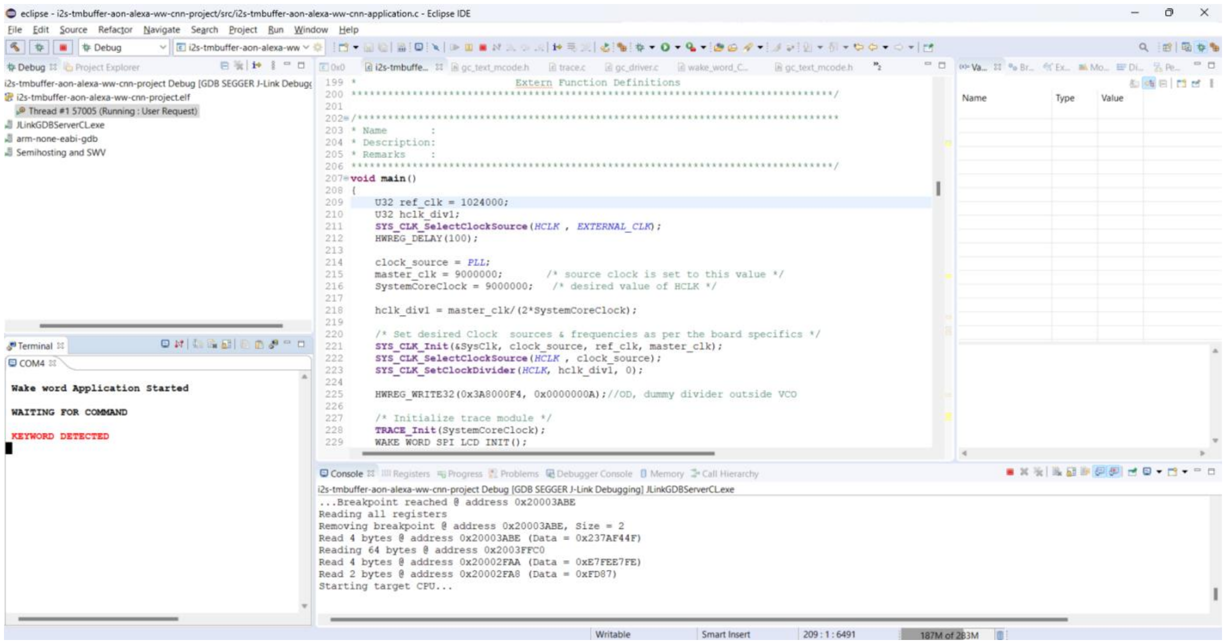
- Serial Terminal will appear, Run the application



- Observe the initial output on serial terminal



- Speak the word Alexa, if the detection happens, the following output appears

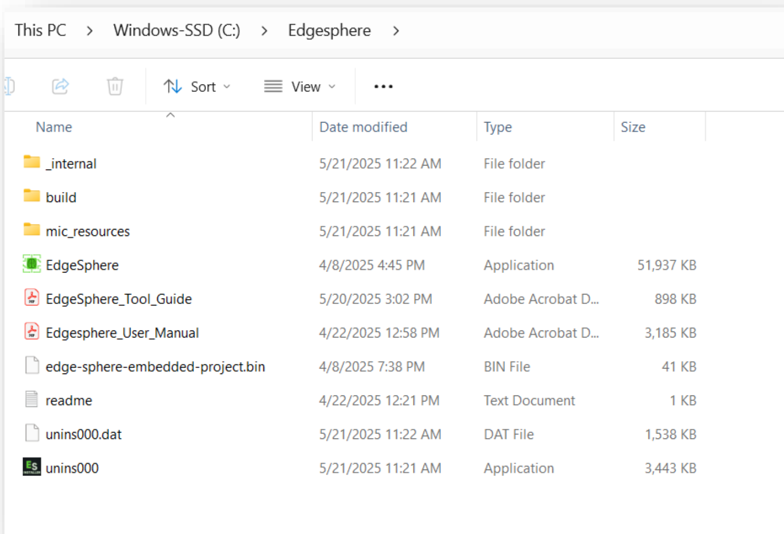


- LED (GPIO_PIN_1) will indicate I2S Mic ON.
- LED (GPIO_PIN_1, GPIO_PIN_2, GPIO_PIN_3) will indicate when keyword will be detected

# 5 Retraining the Model with Custom Wake Word

## 5.1 Collect Audio Samples from DVK Board:

- Navigate to the application directory:
  C:\EdgeSphere and invoke Edge Sphere Application and refer to the document "EdgeSphere_Tool_Guide".




## 5.2 Organize Audio Files:

- You will find the .wav files in the following directory:
  C:\Edgesphere\Collected_Data\Audio
- Create a folder named dataset in:
  C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\training_tool_chain\python
- Inside the dataset folder, create two subfolders:
  - wake-word: Paste the .wav files corresponding to wake word samples (Collected via edge sphere tool).
  - not-wake-word: Paste .wav files of non-wake-word audio samples for training

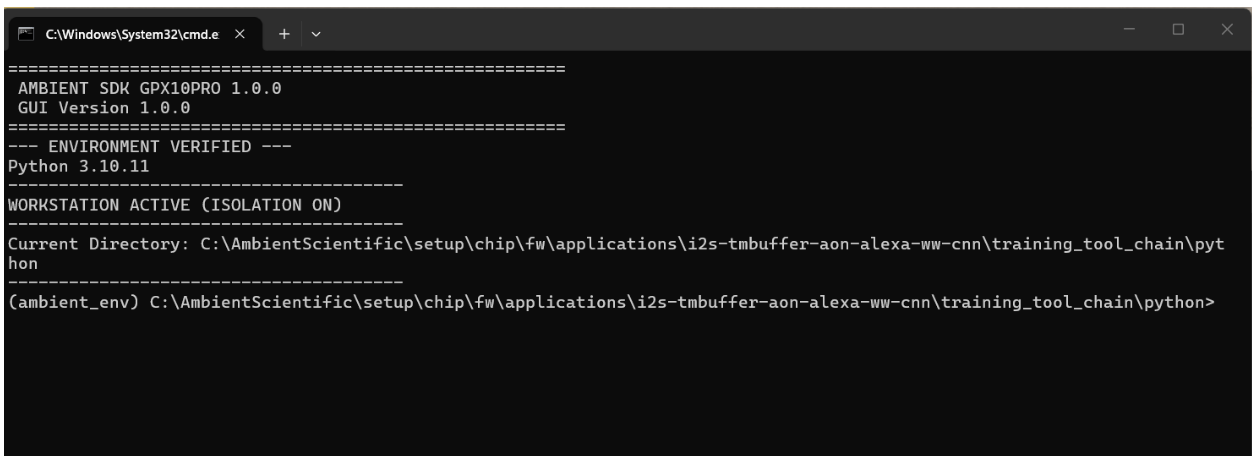
## 5.3 Extract Features from Audio Samples:

Navigate

“C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\training_tool_chain\python”

- Now open cmd prompt


- Run command: **ambient**




- Run the following command to extract features from the audio samples and generate .npy files:
  python Feature-Extraction.py

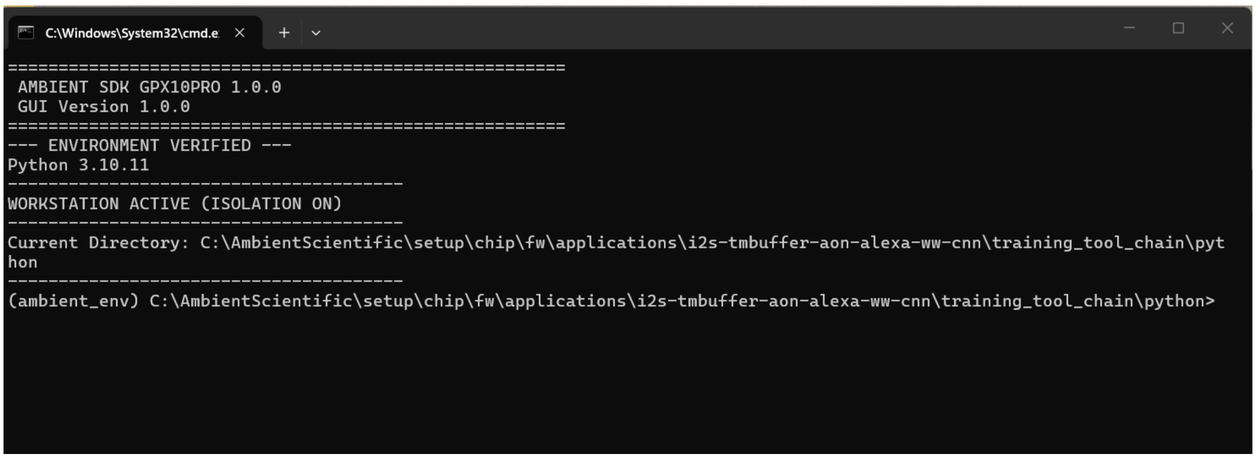
## 5.4 Train the Model:

Navigate

“C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\training_tool_chain\python”

- Now open cmd prompt


- Run command: **ambient**




- Run the following command to train the model:
  python wake-word-model-training.py

- After execution, you will have the following model files:
  - keras model (.h5): wake-word-model.h5
  - TFLite model (.tflite): wake-word-model-quantized.tflite

## 5.5 Changing the Model Configuration

- Save the <model_name...>.h5 here

"C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test"

- Navigate "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test\ModelRefresh.ps1"

  Can change the model name as per the user's requirement.

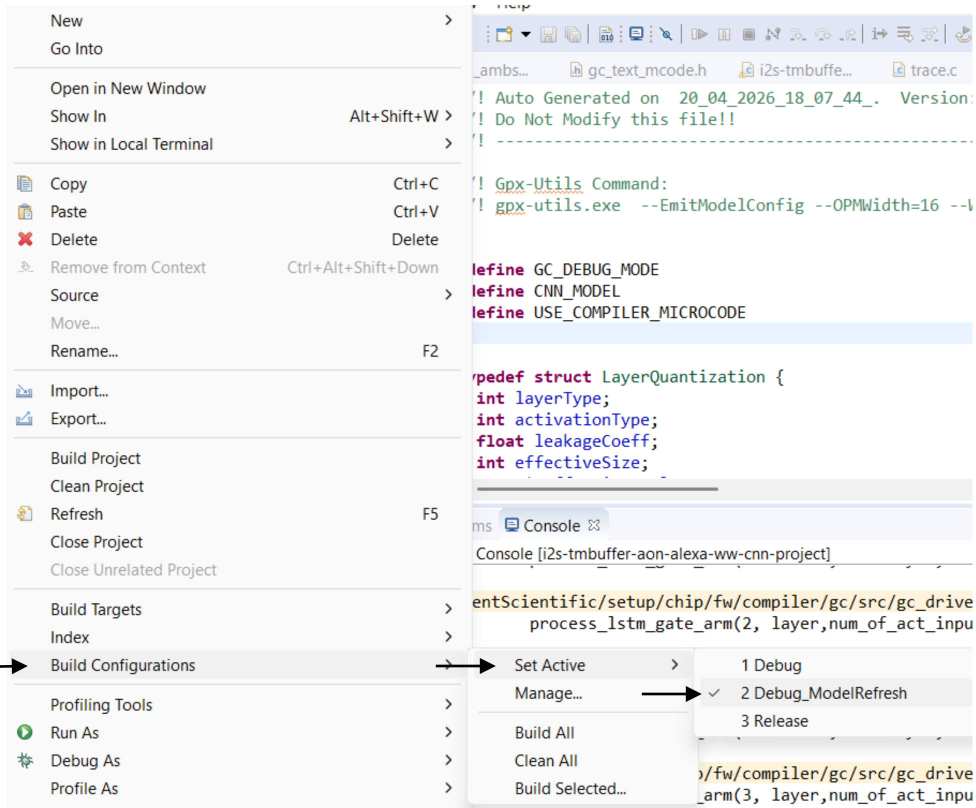
```
# Model refresh for Eclipse Debug_ModelReferesh prebuild.
# Single script: SDK path setup (ambient / ambient_env) + gpxCLITool run.
# Console: minimal status.  model_log.txt: full gpxCLITool output from cmd redirect.
#
# Optional env (or test\local-env.ps1): AMBIENT_SETUP, AMBIENT_PYTHON, AMBIENT_ARM_GCC_BIN

# --- User settings (edit for your project) ---
$modelName = "wake-word-model.h5"
$gpxProfile = "5"                                    # gpxCLITool -p
$modelOutDir = "../i2s-tmbuffer-aon-alexa-ww-cnn/"     # gpxCLITool -d
```

## 5.6 Use the Trained Model:

- Copy the .h5 model to the following directory: C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test
- Go to eclipse
- Right click on project

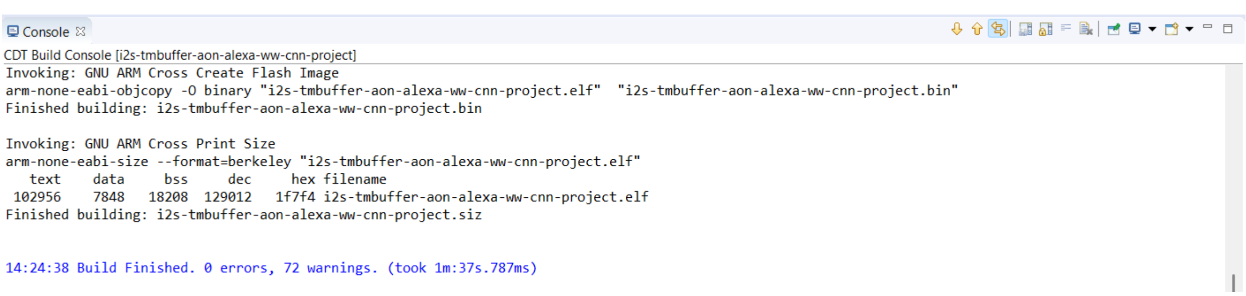
- Go to *Build Configurations and* Select **"Debug_ModelRefresh"** under *"Set Active" Option*




- Perform a **Clean and Build**

**The model name can be verified in the console output.**

- Wait until model deployment to GPX-Hw is complete



- Debug statements will now appear in ***model_log.txt*** You can view them at: "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test\model_log.txt"

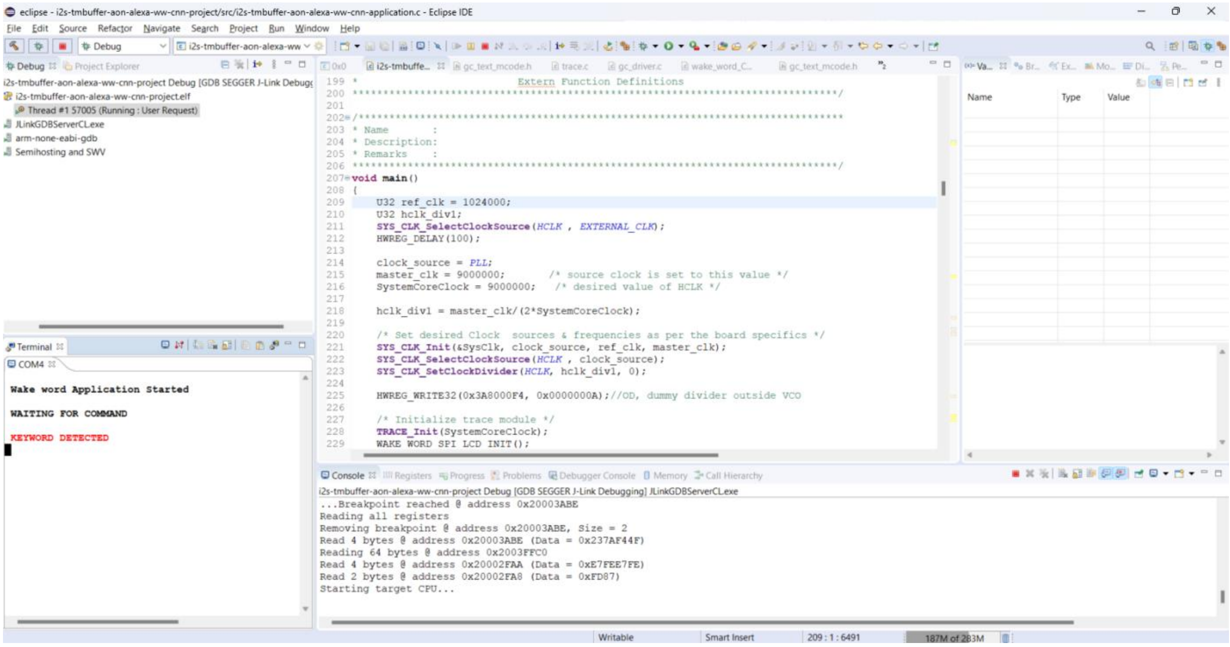
```
⍰[0m⍰[92m      //! BUILD SUCCESS!!. ⍰[0m⍰[93m⍰[45m//! Target image i2s-tmbuffer-aon-alexa-ww-cnn-project.elf successfully generated...

⍰[0m⍰[44m⍰[93m//! RUNNING THE TARGET...
⍰[0m⍰[33m      //! Invoking JLINK GDB_SERVER PROCESS (JLinkGDBServerCL.exe) ...⍰[0m⍰[92m      //! SUCCESS!
⍰[0m⍰[33m      //! Invoking GDB_CLIENT PROCESS ...⍰[0m⍰[92m           //! SUCCESS!
⍰[0m⍰[44m⍰[93m
//! RUNNING THE GENERATED TASK ON TARGET & GETTING RESULTS...
⍰[0m⍰[33m      //! Opening serial terminal with COM-3 PORT,  BAUD_RATE: 9600... ⍰[0m⍰[92m             //! SUCCESS!
⍰[0m⍰[95m//! PRESS 0 to EXIT... Please wait...
⍰[0m⍰[33m      //! Closing Serial COM-3 port ...
⍰[0m⍰[92m              //! SUCCESS!
⍰[0m⍰[33m      //! Invoking GDB_CLIENT PROCESS ...⍰[0m⍰[92m           //! SUCCESS!
⍰[0m⍰[33m      //! Triggered quit command...Please Wait...
⍰[0m⍰[91m//! Gracefully, Terminating the child processes...Please Wait...
⍰[0m⍰[34m****/
⍰[0m
```

- Return to the Eclipse IDE and switch the build configuration, **Debug_ModelRefresh to Debug**

- Find the .elf file in the "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\i2s-tmbuffer-aon-alexa-ww-cnn-project\Debug\i2s-tmbuffer-aon-alexa-ww-cnn-project.elf

- Run the application

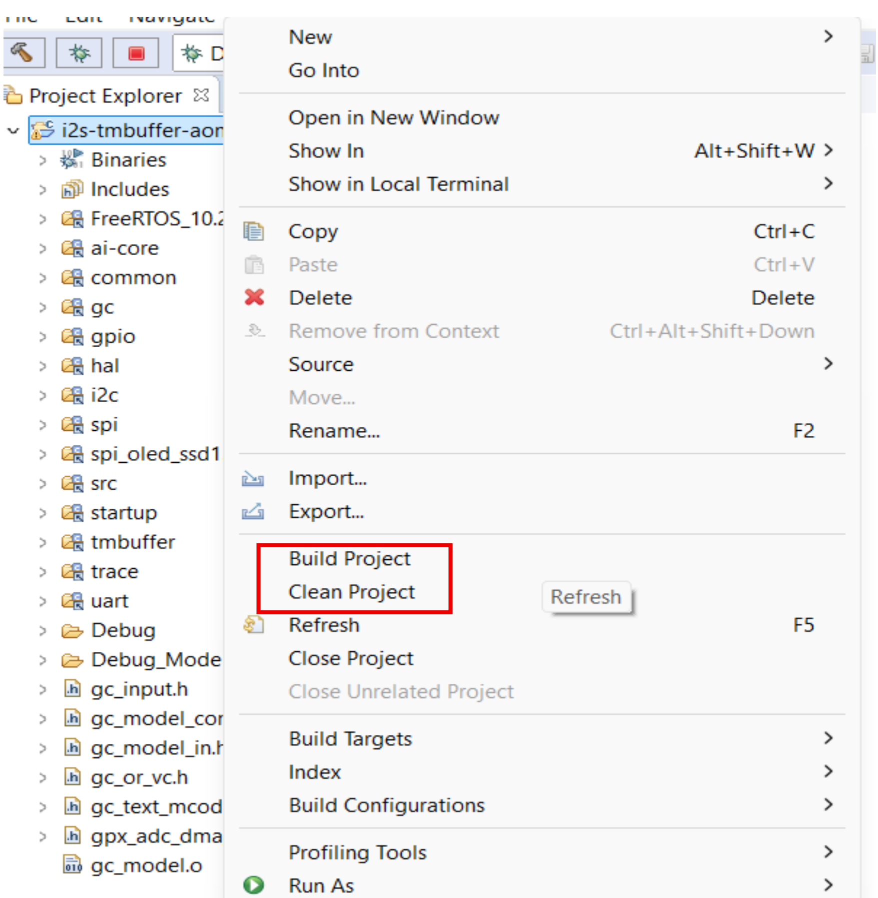


- Speak the word, if the detection happens, the following output appears

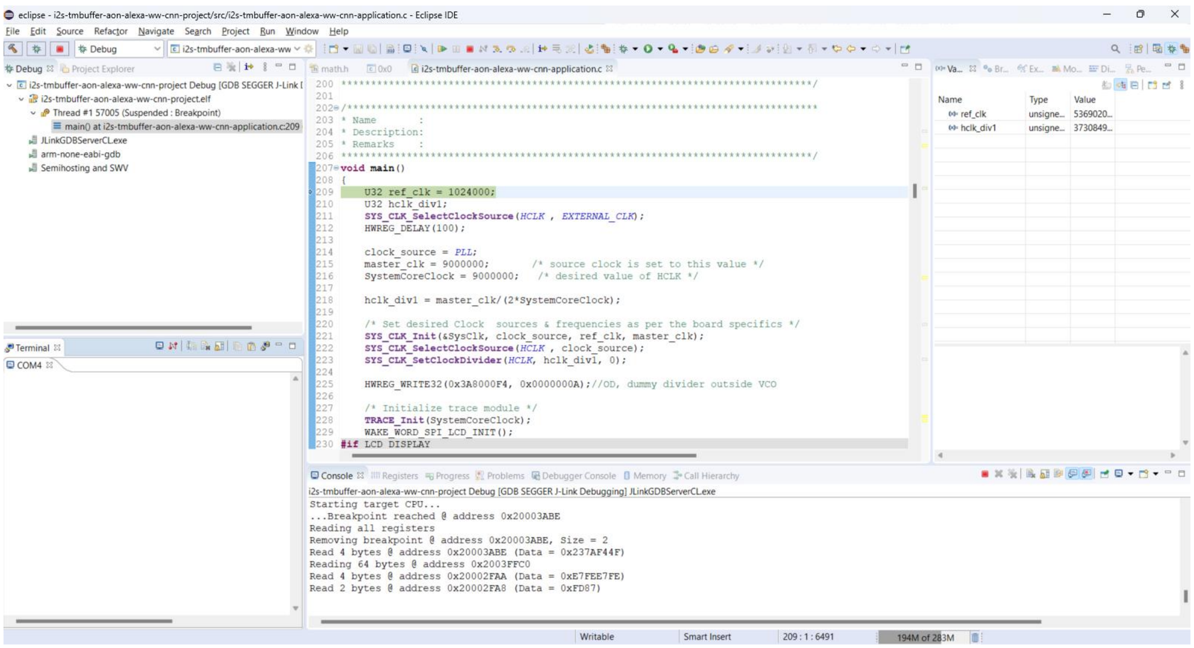
# 6 Build and Run

## 6.1 General Debug

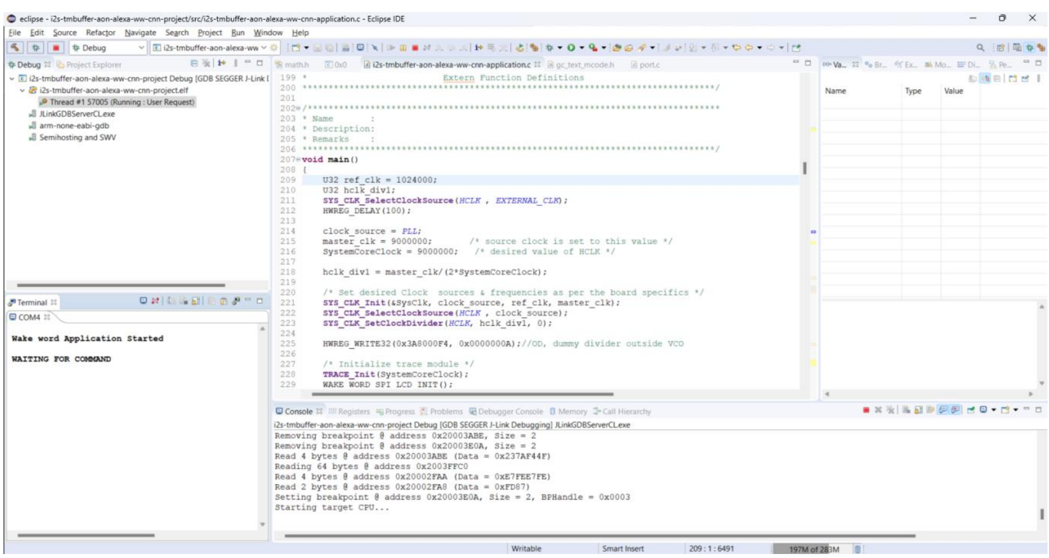
- Import application from here: "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn"
- Right click on project for **Clean and Build**



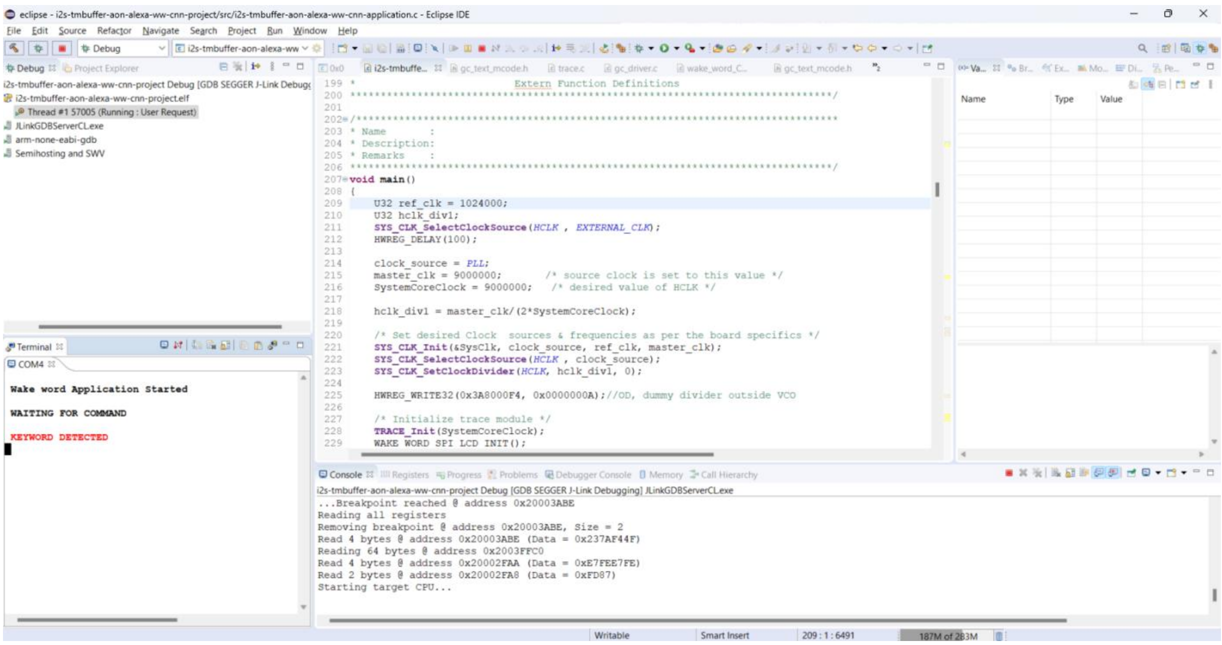
- Serial Terminal will appear, Run the application



- Observe the initial output on serial terminal



- Speak the word Alexa, if the detection happens, the following output appears

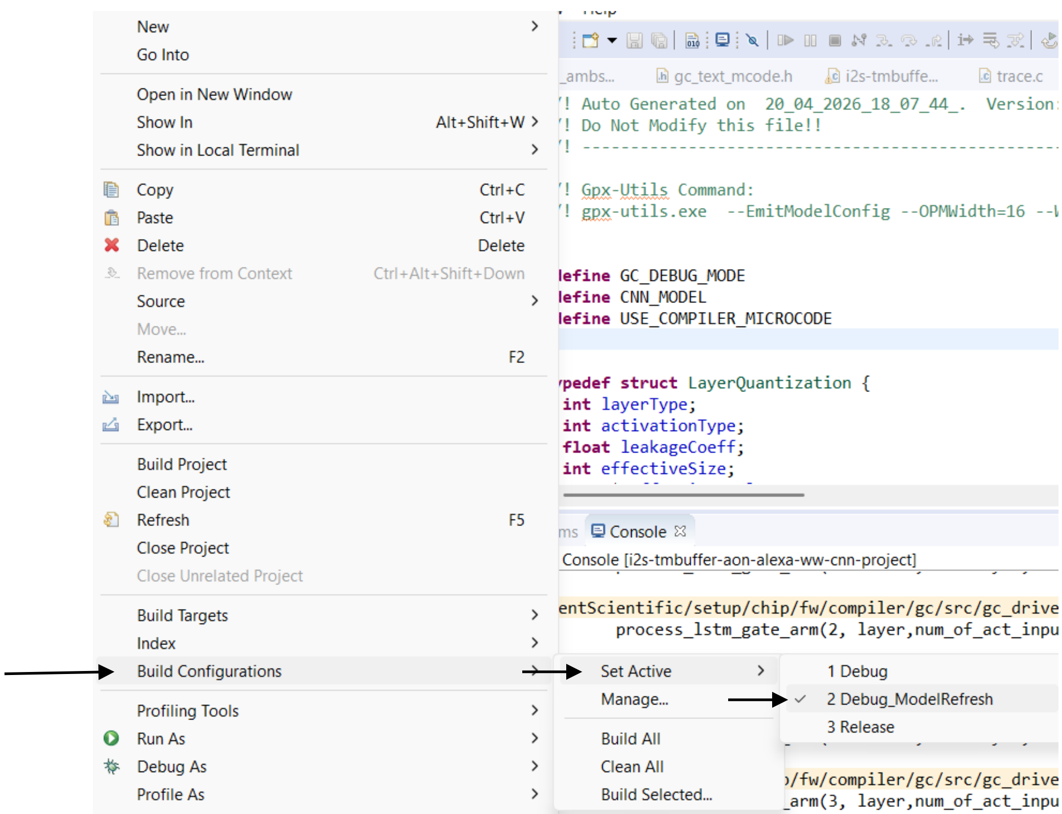



- LED (GPIO_PIN_1) will indicate I2S Mic ON.
- LED (GPIO_PIN_1, GPIO_PIN_2, GPIO_PIN_3) will indicate when keyword will be detected

## 6.2 Change in Model Architecture/Model Rename

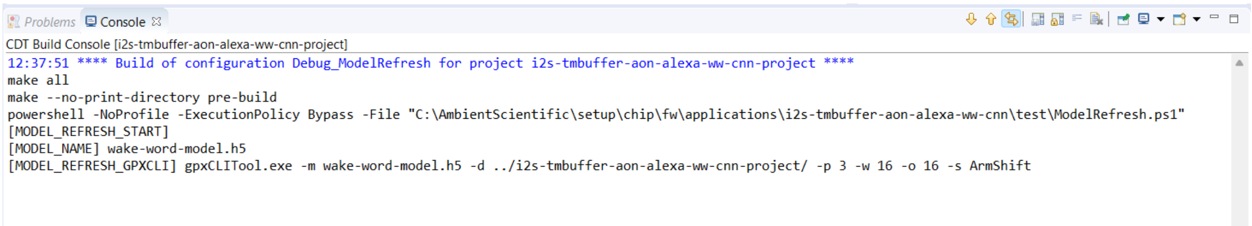
### 6.2.1 Use the Trained Model:

- Copy the .h5 model to the following directory: C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test
- Go to eclipse
- Right click on project
- Go to *Build Configurations and* Select **"Debug_ModelRefresh"** under *"Set Active" Option*



- Perform a **Clean and Build**

**The model name can be verified in the console output.**


Problems Console
CDT Build Console [i2s-tmbuffer-aon-alexa-ww-cnn-project]

```
12:37:51 **** Build of configuration Debug_ModelRefresh for project i2s-tmbuffer-aon-alexa-ww-cnn-project ****
make all
make --no-print-directory pre-build
powershell -NoProfile -ExecutionPolicy Bypass -File "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test\ModelRefresh.ps1"
[MODEL_REFRESH_START]
[MODEL_NAME] wake-word-model.h5
[MODEL_REFRESH_GPXCLI] gpxCLITool.exe -m wake-word-model.h5 -d ../i2s-tmbuffer-aon-alexa-ww-cnn-project/ -p 3 -w 16 -o 16 -s ArmShift
```

- Debug statements will appear in **model_log.txt** for observation navigate to "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test\model_log.txt"

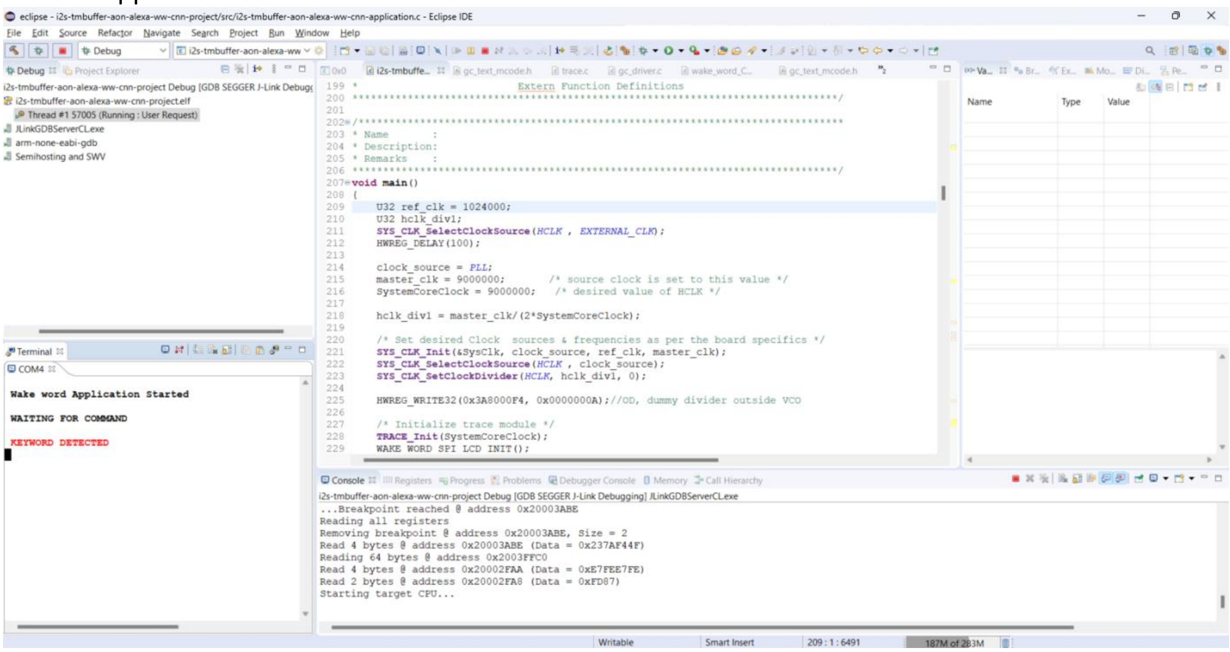
```
⍰[0m⍰[92m      //! BUILD SUCCESS!!. ⍰[0m⍰[93m⍰[45m//! Target image i2s-tmbuffer-aon-alexa-ww-cnn-project.elf successfully generated...

⍰[0m⍰[44m⍰[93m//! RUNNING THE TARGET...
⍰[0m⍰[33m      //! Invoking JLINK GDB_SERVER PROCESS (JLinkGDBServerCL.exe) ...⍰[0m⍰[92m       //! SUCCESS!
⍰[0m⍰[33m      //! Invoking GDB_CLIENT PROCESS ...⍰[0m⍰[92m           //! SUCCESS!
⍰[0m⍰[44m⍰[93m
//! RUNNING THE GENERATED TASK ON TARGET & GETTING RESULTS...
⍰[0m⍰[33m      //! Opening serial terminal with COM-3 PORT,  BAUD_RATE: 9600... ⍰[0m⍰[92m            //! SUCCESS!
⍰[0m⍰[95m//! PRESS 0 to EXIT... Please wait...
⍰[0m⍰[33m      //! Closing Serial COM-3 port ...
⍰[0m⍰[92m              //! SUCCESS!
⍰[0m⍰[33m      //! Invoking GDB_CLIENT PROCESS ...⍰[0m⍰[92m           //! SUCCESS!
⍰[0m⍰[33m      //! Triggered quit command...Please Wait...
⍰[0m⍰[91m//! Gracefully, Terminating the child processes...Please Wait...
⍰[0m⍰[34m****/
⍰[0m
```

- Return to the Eclipse IDE and switch the build configuration, **Debug_ModelRefresh to Debug**

- Find the .elf file in the "C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\i2s-tmbuffer-aon-alexa-ww-cnn-project\Debug\i2s-tmbuffer-aon-alexa-ww-cnn-project.elf

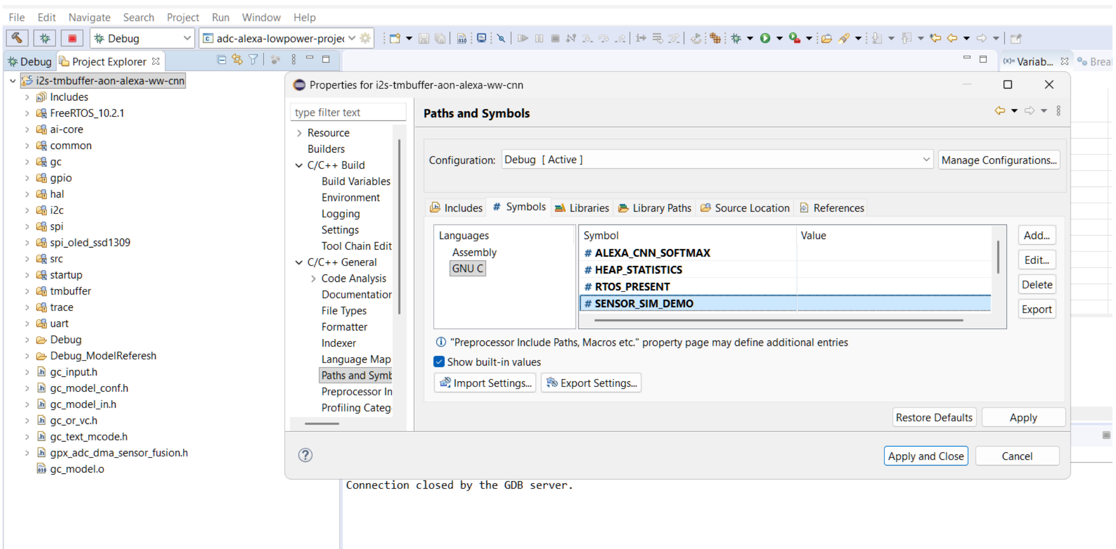
- Run the application




- Speak the word, if the detection happens, the following output appears

## Static mode: (optional)

To run the static data, delete “SENSOR_SIM_DEMO” from project paths and symbols setting as shown below.




This flag will be set to 1 by default when removed “SENSOR_SIM_DEMO” from properties as shown below

```
#ifndef SENSOR_SIM_DEMO
#define ENABLE_WAKE_WORD_STATIC_DATA          1
#else
#define ENABLE_WAKE_WORD_STATIC_DATA          0
#endif
/*****************************************************
```

To load the static data input when running application in static mode (optional) add -i ./input.txt please prefer cmds.txt “C:\AmbientScientific\setup\chip\fw\applications\i2s-tmbuffer-aon-alexa-ww-cnn\test”

```
$toolExe = $paths.ToolExe
$args = "-m $modelName -d $modelOutDir -p $gpxProfile -w 16 -o 16 -s ArmShift"

# gpxCLITool prompts for "0" to continue/end (often after build). Piping (echo 0) only
# feeds stdin at start; use a stdin file so each prompt reads the next "0" automatically.
$stdinFile = Join-Path $paths.WorkDir "gpx_stdin.txt"
$modelLog = Join-Path $paths.WorkDir "model_log.txt"
1..30 | ForEach-Object { "0" } | Set-Content -Path $stdinFile -Encoding ASCII
$cmd = "set ""PATH=$pathPrefix;%PATH%"" && ""$toolExe"" $args < ""$stdinFile"" > ""$modelLog"" 2>&1"
```